

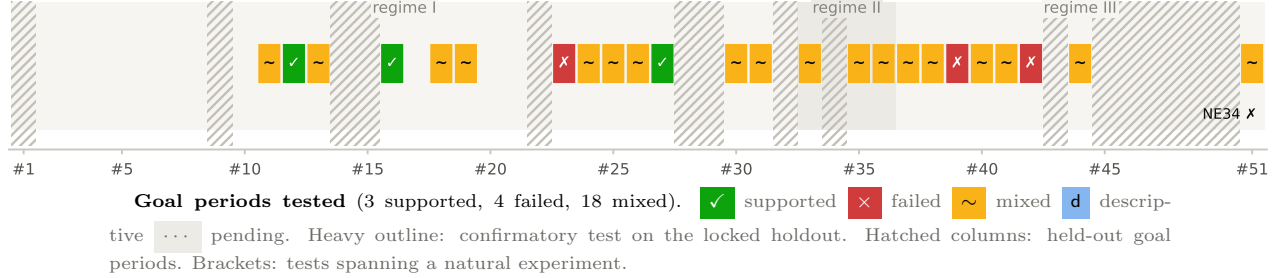
H12: Groupthink is dimensional collapse: few collective modes, shrinking participation ratio at consensus

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 52% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. On corrected data the activity market mode is mostly artifact; talk and content modes are real, a strong field or a contested vote compresses content, kickoffs do not, and low-dimensional weeks are agents restating themselves.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

24 non-holdout units, two arms; **round 1b** re-ran the spin spectra on the corrected activity table (round 1 had lost half of all events) with DQ8's calibrated null, re-checked content and PR with a second embedding model, and added two period-native tests. **Spectrum:** the activity “market mode” of round 1 (22/24 units) mostly disappears: 18/24 under the old null, 7/24 under the calibrated one. It was largely a day-level artifact of the bug. The collective modes that hold are talk (21/24; they now separate rooms in 7/10 two-room units) and content (24/24 in both models). **Dimensionality:** no collapse at 20 kickoffs (robust to model and dedup); the low-PR weeks #38–#40 are agents restating themselves (exact copies 0%). Natively, a strong field *does* compress content: PR is 25% lower while a #12 debate motion is on ($p = 0.01$), and lowest at #26's contested vote. Holdout not run.

PHYSICAL MODEL

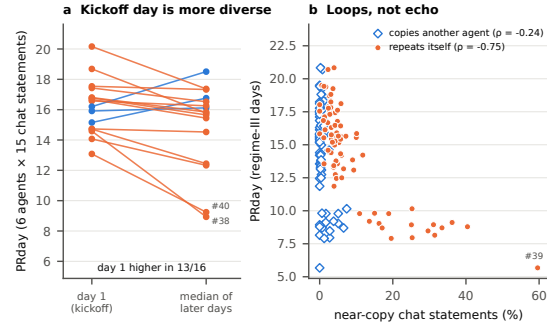
Models 01 (inverse Ising), 11 (vector spins), random-matrix theory. $s_i(t)$: agent i 's ± 1 activity or talk spin per minute, or whitened content vector per 30-min window (N agents, T samples). C : equal-time correlation matrix, eigenvalues λ_k ; λ_+ : noise edge (cross-day surrogates in round 1; trimmed block-shift surrogates in 1b); $u = 1/\sqrt{N}$; $\bar{\rho}$: mean pair correlation. PR: participation ratio of the content covariance Σ ($d = 32$), bias-corrected via $W = \nu S$, $\nu = n - 1$.

$$C_{ij} = \langle s_i s_j \rangle_t, \quad \lambda_+ = (1 + \sqrt{N/T})^2, \quad k = \#\{\lambda_k > \lambda_+\}$$

$$\text{VR} = u^\top C u \leq \lambda_1 \approx 1 + (N-1)\bar{\rho} \quad (\text{iff uniform mode})$$

$$\text{PR} = (\text{tr } \Sigma)^2 / \text{tr } \Sigma^2 = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$$

$$\mathbb{E}(\text{tr } W)^2 = \nu^2 a + 2\nu b, \quad \mathbb{E} \text{tr } W^2 = \nu a + \nu(\nu+1)b \Rightarrow \hat{a}, \hat{b}$$



Observables. Round 1 (PR inputs unchanged in 1b). (a) Agent-balanced PR (6 agents $\times$ 15 chat statements) on the kickoff day vs the median of later days, 16 periods with $N \geq 10$: day 1 is higher in 13 (orange); with the second embedding model 11/16, after removing restatements 9/15. (b) Regime-III days: PR falls as agents restate their own earlier messages ($\rho = -0.75$; -0.76 with DQ5's either-model flag), not other agents' ($\rho = -0.24$). The low cluster is #38–#40, robust to both models.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The portable results are about what is *not* collective. The activity “market mode” was mostly a data bug plus the daily schedule: under a calibrated null agents are not busy or idle together beyond chance in most weeks. What agents talk about does co-move (content and talk modes), and a strong uniform field (a debate motion) or a contested vote measurably narrows it, while goal kickoffs widen it. The cheapest useful monitor remains self-restatement: it flags agents stuck rephrasing themselves, the village's real low-dimensional states. The groupthink alarm stays unvalidated: cross-agent echo never exceeded a few percent.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict
P1 few activity modes (cross-day)	$k=1$ in 22 → 18/24; $k=0$ in 2 → 6	×
DQ8 null (trim, block activity mode in 7/24; talk 21/24 shift)		new
P1' beyond lulls	14 → 8/24 (lulls now rare)	×
P2 Curie–Weiss shape; III > I	uniform 22/22 → 16/18; λ_1 /edge III 1.62 → 1.13	×
P3 talk modes separate rooms	3/10 → 7/10; but $k_{\text{talk}} > k_{\text{act}}$ often	×
P4 content modes	24/24; gte 24/24 (ρ 0.99)	core ✓
P6 kickoff collapse	+7%; bge +4%, gte +9% ($p \geq 0.77$)	×
P7 day 1 vs later (R4)	13/16 higher; gte 11/16; dedup 9/15	×
G12 motion field (native)	PR on < off in 7/9, -25%, $p = 0.010$	✓
G26 vote instants (native)	contested vote pct 0.07; re-election ~ 0.37	~

GOAL PERIODS IN DETAIL

G11 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G12 ✓ supported
 G13 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G16 ✓ 1b: supported (round 1: supported; corrected activity...
 G18 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G19 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G23 × 1b: failed (round 1: mixed; corrected activity table,...
 G24 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G25 ~ 1b: mixed (round 1: mixed; corrected activity table, ...
 G26 ~ 1b: mixed (replication, round 1: mixed; corrected act...

+15 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The activity-mode collapse is explained by the bug's day-level common loss plus the scheduler's start/stop (H50 finds the latter independently from call windows); within-block co-activation rose on the fixed table. PR measures what agents write, including format: debate turns are structured speeches, and the #12 off window is short. P7's direction depends on the embedding model and on restatement removal. Whitening is per regime; PR levels compare only within a regime. Small units: 3-day units have ~14% false positives under the cross-day edge. The holdout is untouched; Amendment 2's C1–C3 (activity mode) and C5 (day-1 expansion) look unlikely to pass as frozen.

ROUND 2

Re-freeze the confirmatory C1–C3 on talk and content modes with the trimmed block-shift edge before any holdout run; repeat the #12 field test on other field switches (verdicts in H21, operator corrections); add a restatement-rate monitor with the either-model flag.

- **Essence.** Loops are fixed points of an agent's self-conditioned generation.
- **H12-R1** Loop onset is predicted by the share of an agent's context filled with its own recent output; crossing a threshold is a bifurcation to a fixed point (E2).
- **H12-R2** Only novel external input breaks a loop, and the escape probability scales with the input's novelty (embedding distance), not with who sent it.
- **H12-R3** Swarm groupthink never forms because each agent is its own echo chamber: self-echo dominates cross-echo.

Second figure. (a) Top eigenvalue over the 95% surrogate edge per scored unit: activity under round 1 (gray) vs the corrected table (blue) and the calibrated DQ8 edge (red); talk under the DQ8 edge (green) stays above 1 in 21/24. (b) #12: participation ratio after each verdict vs while the motion is on, paired by debate, in two embedding models.

